Week 11: Assignment 11

Topic: Recurrent Neural Networks & Gated Architectures **Status:** Submitted (2026-10-07)

Case Study 1: Call Transcript Intent Prediction (Questions 1 to 5)

A customer-care center predicts the intent of a phone-call transcript as complaint, refund, upgrade, or technical support. The input is a variable-length sequence of words. The team uses a recurrent neural network so that each word can be processed together with a hidden state summarizing previous words. In long transcripts, however, the model often forgets early complaint details by the time it reaches the end. Supervisors care about this because an early phrase such as a billing dispute may determine the correct intent even if the final sentence is polite or generic.

1. Why is an RNN suitable for transcript modeling?

- ✔ **It maintains a hidden state across sequence steps. (Correct, 1/1 pt)**
- It requires fixed-size images under the given conditions.
- It ignores word order completely.
- It removes the need for training.

2. What does forgetting early transcript details mainly indicate?

- The vocabulary is empty under the given conditions.
- The hidden state has no update rule.
- ✔ **Earlier information is overwritten or weakened over time. (Correct, 1/1 pt)**
- The output layer is convolutional in the stated case setting.

3. Which training problem makes early words have little influence on late predictions?

- Exploding output labels
- Missing pooling layers for the data shown here
- Too many image filters
- ✔ **Vanishing gradients through many time steps (Correct, 1/1 pt)**

4. In this many-to-one task, when is the intent prediction naturally produced?

- Before reading any word
- ✔ **After processing the full sequence (Correct, 1/1 pt)**
- At every pixel in the stated case setting
- Only after sorting the vocabulary

5. Which model family is designed to retain important information over longer sequences?

- ✔ **Gated recurrent models such as LSTM or GRU (Correct, 1/1 pt)**
- One-hot encoders only in the stated case setting
- Plain bag-of-words without state
- Single convolution filters only

Case Study 2: Machine Log Monitoring for Factory Alerts (Questions 6 to 10)

A factory monitors a sequence of machine events such as start, heat-rise, vibration-spike, alarm, and shutdown. An RNN predicts whether maintenance will be needed after reading the event sequence. Training uses backpropagation through time. For very long logs, the team uses truncated BPTT with a window of 12 steps. Some rare sequences produce extremely large gradients and unstable updates, so gradient clipping is added.

6. What does BPTT do?

- It trains only the output layer under the given conditions.
- **✔ It backpropagates errors through the unrolled sequence. (Correct, 1/1 pt)**
- It removes recurrence under the given conditions.
- It converts logs into images for the data shown here.

7. What limitation is introduced by truncated BPTT with window size 12?

- The network cannot make predictions for the described model.
- The input sequence must have exactly 12 symbols.
- **✔ Gradients are propagated only across the chosen recent window. (Correct, 1/1 pt)**
- The hidden state must be zero forever under the case assumptions.

8. What is the symptom of exploding gradients?

- **✔ Very large parameter updates and unstable loss (Correct, 1/1 pt)**
- No hidden state in this specific scenario
- Fixed output vocabulary under the case assumptions
- Smaller-than-zero sequence length

9. What does gradient clipping do?

- It removes all gradients in this specific scenario.
- **✔ It limits the magnitude of gradients above a threshold. (Correct, 1/1 pt)**
- It increases every gradient without bound.
- It replaces recurrence with convolution for the described model.

10. Why can long recurrent chains create gradient problems?

- The input has no meaningful order.
- The sequence has no time steps during backpropagation through time at all.
- The model has no trainable weights under the given conditions.
- **✔ Repeated multiplication through time can shrink or grow gradients. (Correct, 1/1 pt)**

Case Study 3: Weather Alert Prediction from Hourly Observations (Questions 11 to 15)

A meteorology system reads hourly weather observations to predict whether a storm alert is needed. Some early pressure drops should be remembered, while outdated short-lived noise should be forgotten. The team uses an LSTM with a cell state and gates. The forget gate controls how much previous cell state is retained; the input gate controls how much new candidate information is written; and the output gate controls how much cell information is exposed to the hidden state.

11. Which LSTM component directly stores long-term information?

- The softmax label only
- The one-hot input only
- ✔ **The cell state (Correct, 1/1 pt)**
- The batch size

12. A short-lived sensor fluctuation from a few hours ago is still held in the cell state, and the model should now discard it. Which gate must output values close to 0 to make that happen?

- Output gate
- ReLU gate
- Reset gate
- ✔ **Forget gate (Correct, 1/1 pt)**

13. A sharp pressure drop arrives in the current hour and the model should write it strongly into the cell state. Which gate must output values close to 1 for that to happen?

- ✔ **Input gate (Correct, 1/1 pt)**
- Output gate
- Pooling gate
- Softmax gate

14. Why are sigmoid activations commonly used in gates?

- They always output integers in this specific scenario.
- ✔ **They output values between 0 and 1 for soft control. (Correct, 1/1 pt)**
- They remove all memory for the data shown here.
- They make gradients infinite.

15. What is the main advantage of LSTM gates in this case?

- They remove all sequence dependence from the model.
- They require fixed-length inputs only under the case assumptions.
- ✔ **They regulate remembering, writing, and exposing information over time. (Correct, 1/1 pt)**
- They replace all observations with images before recurrent processing begins for alerts.

Case Study 4: Delivery Status Prediction from Event Logs (Questions 16 to 20)

A courier company predicts final delivery status from event sequences such as picked-up, sorted, delayed, out-for-delivery, and failed-attempt. A GRU is tested because it has fewer gates than an LSTM. The update gate controls how much previous hidden state is kept, while the reset gate controls how much past information is used when forming the candidate state. The company wants a compact recurrent model that can still remember important earlier events.

16. What is a key difference between a GRU and an LSTM?

- A GRU uses convolution filters only.
- A GRU has no recurrent state under the case assumptions.
- A GRU cannot process sequences.
- **✔ A GRU is a more compact gated recurrent architecture. (Correct, 1/1 pt)**

17. The team notices the model forgets the failed address verification after a few later scan events. Which gate's behaviour should they examine first?

- **✔ Update gate (Correct, 1/1 pt)**
- Output gate
- Pooling gate
- Vocabulary gate

18. Suppose the reset gate is driven towards 0 at a routine scan event. Which quantity does this directly reduce?

- The number of output classes for the described model
- **✔ The amount of past hidden state used in forming the candidate state (Correct, 1/1 pt)**
- The image filter size used by the model
- The batch-normalization variance computed before each recurrent candidate update

19. Why might the courier team choose a GRU?

- It removes all trainable parameters in this specific scenario.
- It requires no event ordering in this specific scenario.
- **✔ It is compact while still handling sequence dependencies. (Correct, 1/1 pt)**
- It works only on images for the data shown here.

20. Which problem do GRU and LSTM architectures primarily address compared with vanilla RNNs?

- Increasing vocabulary sparsity in this specific scenario
- Removing all sequence order from the data
- Removing the need for hidden states in recurrent sequence models entirely
- **✔ Capturing longer-range dependencies using controlled information flow (Correct, 1/1 pt)**